

set of possible states

belief (probabilities)

after a move

union of the reachable states



$$\overline{\text{bel}}(x_t) = \sum_{x_{t-1}} p(x_t \mid u_t, x_{t-1}) \text{bel}(x_{t-1})$$

law of total probability

after a reading

keep only the states that fit



$$\text{bel}(x_t) = \eta p(z_t \mid x_t) \overline{\text{bel}}(x_t)$$

Bayes' theorem